

Kirkland's Design Document | TARGET SHIP DATE: 2025-12-22

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PROJECT: CATT

This game is an RPG where the user plays the role of a wandering cat who finds joy in connecting with other cats. After logging in (and reading the backstory), you can click on different cats on the game screen to interact with them. By interacting with different cats, the user is able to gain affection points and raise affection levels. When exchanging dialogue with cats, the user can also encounter fun options, like trivia or jokes. Utilizing the weather API, shown as an icon on the interface, the user can encounter cats of different regions. Encounter all the cats to get a prize!!

FEF: BOOTSTRAP

COMPONENTS

- Sessions
 - Register
 - Takes in username and password as long as the username isn't taken
 - Login/Logout
 - Takes in username and password
 - Accesses database of usernames and passwords for each user to log in
 - Use sessions to keep a user logged in and only log them out if they press a logout button
 - User doesn't have to be logged in to use our site, but progress will not be saved
- Webpages
 - Startscreen
 - <click to start>
 - Gives a prompt for logging in or signing up if not yet logged in
 - Brief lore/backstory to the cat and the game
 - Black screen, just the character in the middle, text box on the bottom

Legend
Italic = optional / additional features
Underline = keywords / important

- Play button also acts as a skip button if you've already seen it and want to skip it
- Weather based encounter maps
 - Webpage templates will be made completely out of placeholders/variables (mainly jinja)
 - Weather + icon will be displayed in the top left corner of the screen
 - Access database of content based on the weather and display it
 - Cats that appear will be based off the energy level set for the weather map
 - Different weather = different cats!!!!
 - Cats will be buttons
 - Click on cat to get an encounter
- Encounters
 - Your cat sprite and the other cat
 - Click on other cat to begin dialogue
 - Tell jokes, ask trivia questions
 - Answering right for trivia or clicking the right response will give you more affection points from the cats
 - *Will have a level system so points are of significance to the player, perhaps a multiplier applied to future interactions*
 - Encountering all the cats will display a button in the profile page where you can click to get the prize
- Profile page
 - See all the cats you've encountered as cards with images, affection points, and level
 - Every fifty affection points you level up
 - Bootstrap progress bar for number of cat encounters
 - Cheat button with password to get all encounters
 - Get final prize button (with a Bootstrap popup of "are you sure")
- Database (SQL)
 - Stores all webpages and data
 - DATABASE TABLE FOR USER INFO:

USER_PROFILE

TEXT	username	PK NOT NULL
TEXT	password	NOT NULL
TEXT	sprite	

- Sprite: profile picture of the account

- DATABASE TABLE FOR ENCOUNTER MAPS:

ENCOUNTER_MAPS		
TEXT	background	
INTEGER	num_cats	
INTEGER	energy_lvl	
TEXT	time	
TEXT	weather	
INTEGER	id	PK

- DATABASE TABLE FOR USER ENCOUNTERS:

USER_ENCOUNTERS		
TEXT	username	
TEXT	cat	
INTEGER	affection	
INTEGER	level	

- DATABASE TABLE FOR ENCOUNTER DIALOGUE:

DIALOGUE		
TEXT	encounter_type	PK
TEXT	response1	
TEXT	response2	
TEXT	response3	

TEXT	response4	
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■ DATABASE TABLE FOR JOKES:

JOKES		
TEXT	category	
TEXT	joke	PK
INTEGER	difficulty	

■ DATABASE TABLE FOR TRIVIA:

TRIVIA		
TEXT	difficulty	
TEXT	answer1	
TEXT	answer2	
TEXT	answer3	
TEXT	answer4	
TEXT	correct_answer	

■ DATABASE TABLE FOR CATS:

CATS		
TEXT	breed	PK
INTEGER	energy_lvl	
INTEGER	difficulty	
TEXT	difficulty2	
INTEGER	response_type	

- Response_type = 0 for jokes
- Response_type = 1 for trivia
- Difficulty: used for jokes (bc it accepts int)
- Difficulty2: used for trivia (bc it accepts str)

- HTML templates

- Login/Register
 - Takes user input via forms
 - Returns different invalid reasons
 - Usernames are **case-insensitive** bc our text capitalizes all letters
- Logout page
- Start page
 - Introduces background of game
 - Start screen for game
 - Links to login/register
- Encounter page
 - Template for every cat maps
 - Cats on the map can range from 1-5 depending on the weather of the location on the top right of screen
- Weather encounters page
 - Takes user input via button clicks for dialogue
 - Must return data and update content upon click of play/continue button
- Profile page
 - Allows you to change profile pictures
 - Takes user input for cheat password
 - Takes user input for changing profile picture
- Settings page
 - Change password feature
- APIs
 - [Cat API](#)
 - Will get cat breeds and energy levels
 - Cats provided randomized through use of the API
 - [Joke API](#)
 - Difficulty can vary based on breed of cat
 - Some cats say very mean jokes
 - [Pirate Weather API](#)
 - Get weather based on location (configurable)
 - [Trivia API](#)
 - Difficulty can vary based on breed of cat
 - [Evil Insults API](#)
- FEF
 - [Bootstrap](#)

- Responsive features & formatting system
 - Badge heading, when encountering a new cat
 - Cards for cats you've already encountered on profile page
 - Progress bar for progress of current encountered cats out of total possible cats
 - Popups for added affection after encounters
- Form styling, for use in login/register
 - Floating labels, validation for login/register page!!
- Navbar functionality

FEF Site Map

Startscreen (PRIVATE)

- | └ main game interface, where you start playing the game and can
- | access other features
- | └ redirect to Login/Register if not yet logged in

└ Login (PUBLIC)

- | └ redirects to Startscreen to introduce background

└ Register (PUBLIC)

- | └ redirects to Startscreen to introduce background

└ Profile (PRIVATE)

- | └ includes all data of user, including username, collection
- | progress, profile picture, etc.

└ Encounters (PRIVATE)

- | └ encounter page where player can click cats on screen to have
- | encounters (randomly generated cats)

| └ Weather Encounter (PRIVATE)

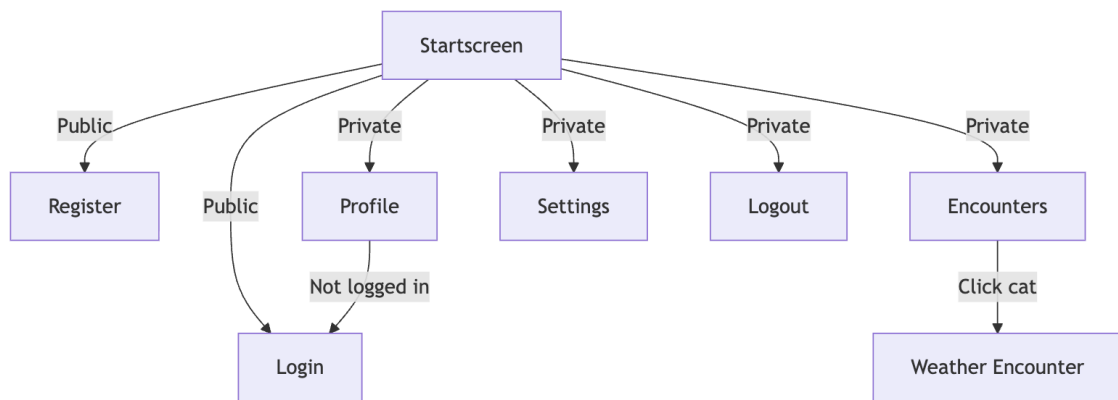
- | └ utilizes the weather API's energy level (within a
- | specific range of temperatures) to randomly select a cat

└ Settings (PRIVATE)

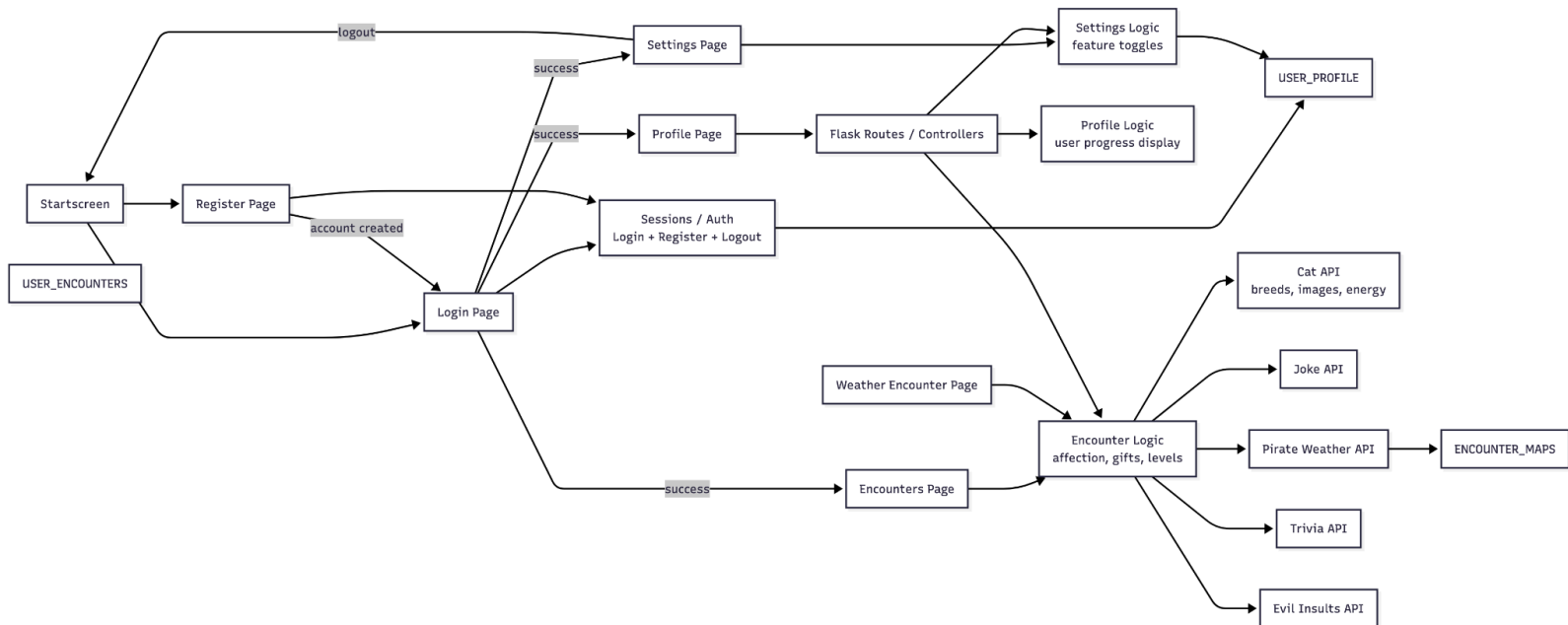
- | └ page to change account password

└ Logout (PRIVATE)

- | └ links to Login page



Component Map



Task Assignments

<u>Task</u>	<u>Role</u>	<u>Agent</u>	<u>Completed?</u>
Bootstrap + HTML templates	Create HTML templates for webpages and add Bootstrap features	Rohan	Y/N
SQLite / Encounters	Integrating APIs with webpage and pulling data from API to insert into data tables	Carrie	Y/N
Flask	Setup routes for webpages	Amy	Y/N
Profile page	Create page and profile features linking to data tables	Rohan, Amy	Y/N
<i>Settings</i>	<i>Create settings page and edit Python app to utilize various settings</i>	<i>ALL</i>	<i>Y/N</i>
<i>CSS</i>	<i>Beautify the cats</i>	<i>ALL</i>	<i>Y/N</i>